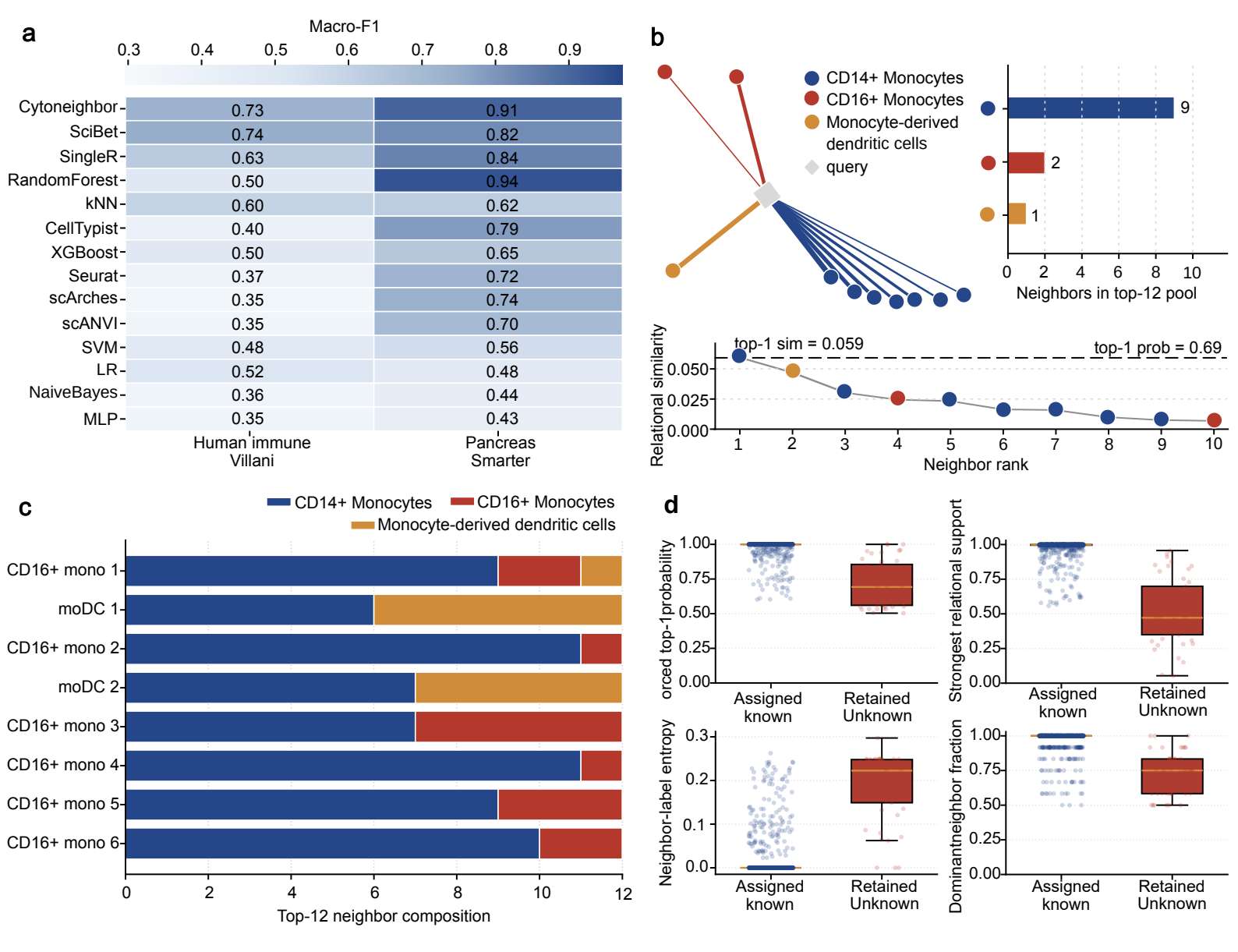

a
Macro-F1
0.3
0.4
0.5
0.6
0.7
0.8
0.9
Cytoneighbor 0.73 0.91
SciBet 0.74 0.82
SingleR 0.63 0.84
RandomForest 0.50 0.94
kNN 0.60 0.62
CellTypist 0.40 0.79
XGBoost 0.50 0.65
Seurat 0.37 0.72
scArches 0.35 0.74
scANVI 0.35 0.70
SVM 0.48 0.56
LR 0.52 0.48
NaiveBayes 0.36 0.44
MLP 0.35 0.43
Human immune Villani
Pancreas Smarter
b
CD14+ Monocytes
CD16+ Monocytes
Monocyte-derived dendritic cells
query
9
2
1
0 2 4 6 8 10
Neighbors in top-12 pool
top-1 sim = 0.059
top-1 prob = 0.69
Relational similarity
0.050
0.025
0.000
1 2 3 4 5 6 7 8 9 10
Neighbor rank
c
CD14+ Monocytes
CD16+ Monocytes
Monocyte-derived dendritic cells
CD16+ mono 1
moDC 1
CD16+ mono 2
moDC 2
CD16+ mono 3
CD16+ mono 4
CD16+ mono 5
CD16+ mono 6
0 2 4 6 8 10 12
Top-12 neighbor composition
d
orced top-1probability
1.00
0.75
0.50
0.25
0.00
Assigned known
Retained Unknown
Strongest relational support
Neighbor-label entropy
0.3
0.2
0.1
0.0
Dominantneighbor fraction